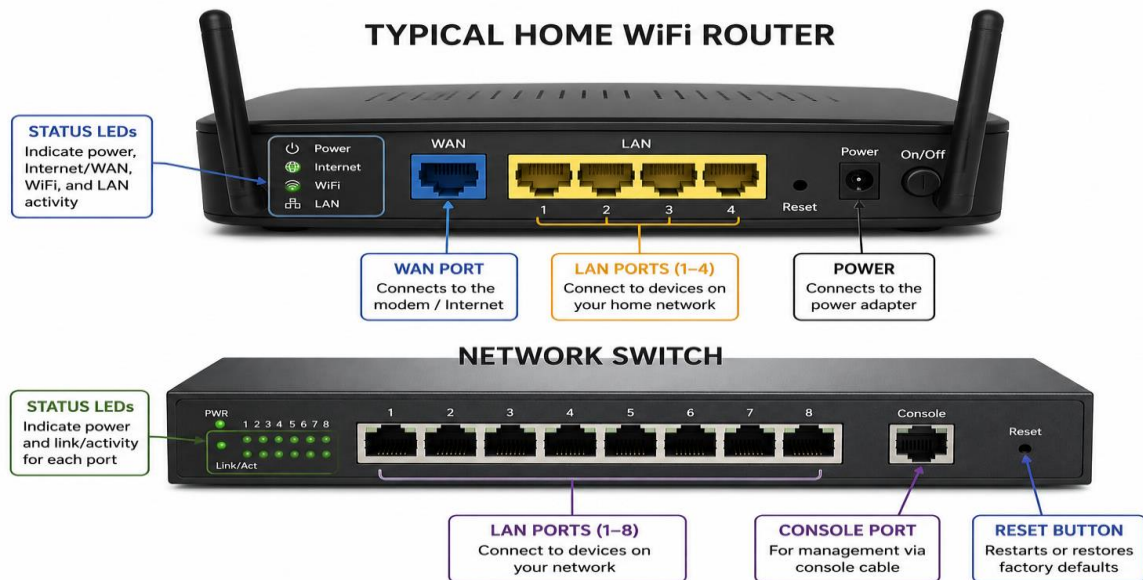


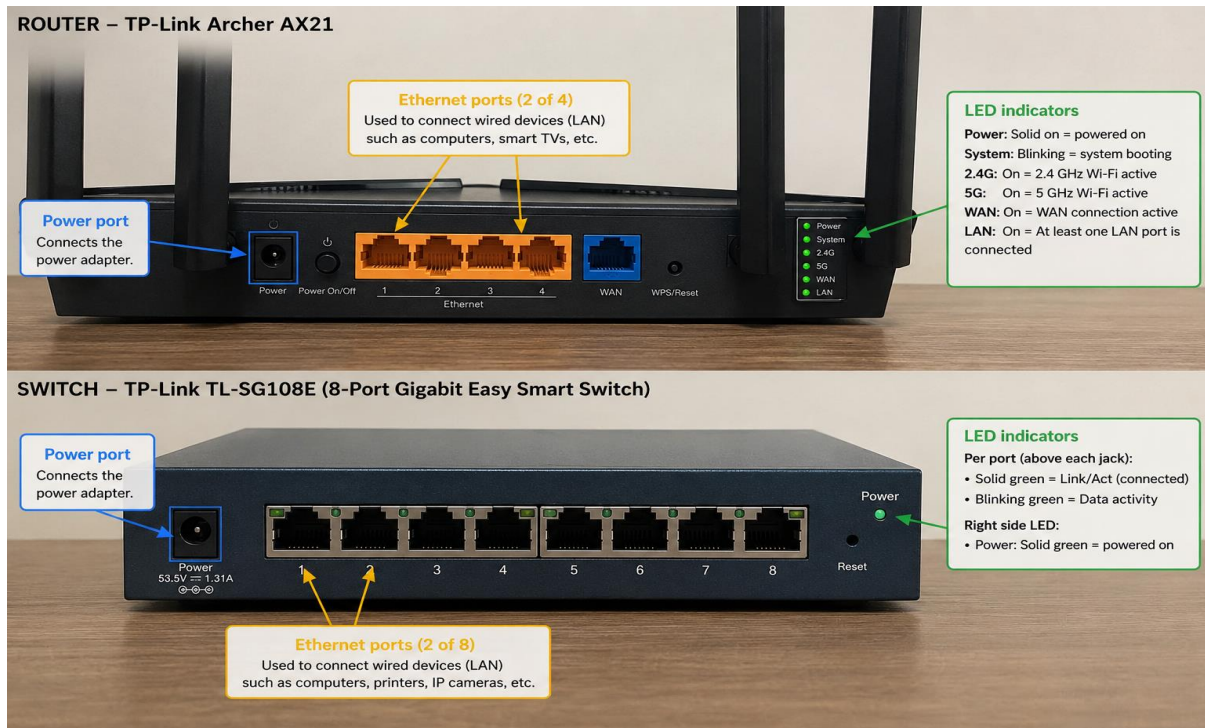
1. Draw a labelled diagram of a typical home Wi-Fi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs.



2. List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic. Think about how each device deals with IP addresses, MAC addresses, and the types of networks they connect.

Differences	Router	Switch
Main Function	Connects different networks (home network to the internet).	Connect devices within the same Local Area Network(LAN).
Traffic Handling	Use IP address to determine the best path and forward data between networks.	Use MAC addresses to forward data directly to the correct device within the LAN.
Network Scope	Routes traffic between separate and can manage internet access.	Transfers traffic within a single network, improving communication between connected devices.

3. Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.



4. Imagine you are setting up a small gaming cafe network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.
- To set up a small gaming café for IPL streaming and multiplayer gaming, I would connect the router first to the ISP line. The router acts as the gateway between the internet and the local network. It receives the internet connection from the ISP, assign IP address to devices and provides security through features like NAT and a firewall.
 - Next, I would connect an 8-port or larger network switch to one of the router's LAN ports. Then, I would connect all 8 gaming PCs to the switch using Ethernet cables.

Justification

- **Router first:** Connects to the ISP, shares the internet connection, manages IP addresses, and protects the network.

- **Switch second:** Expands the number of available LAN ports and allow all 8 gaming PCs to communicate efficiently at high speed.
- **Wired Ethernet connections:** Provide low latency and stable performance. Which is essential for smooth IPL streaming and multiplayer gaming.